MATH256
Linear Algebra HW#6

1. (5.1.6) Is $\begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix}$ an eigenvector of $\begin{bmatrix} 3 & 6 & 7 \\ 3 & 2 & 7 \\ 5 & 6 & 4 \end{bmatrix}$? If so , find the eigenvalue.

2. (5.1.8) Is $\lambda = 1$ an eigenvalue of $\begin{bmatrix} 4 & -2 & 3 \\ 0 & -1 & 3 \\ -1 & 2 & -2 \end{bmatrix}$? If so, find one correnponding eigenvector.

In questions 3 and 4 , find the basis for the eigenspace corresponding to each listed eigenvalue.

3. (5.1.14)

$\begin{bmatrix} 4 & 0 & -1 \\ 3 & 0 & 3 \\ 2 & -2 & 5 \end{bmatrix}$, $\lambda = 3$

4. (5.1.16)

$\begin{bmatrix} 5 & 0 & -1 & 0 \\ 1 & 3 & 0 & 0 \\ 2 & -1 & 3 & 0 \\ 4 & -2 & -2 & 4 \end{bmatrix}$, $\lambda = 4$

5. (5.1.22) A is an n x n matrix . Matrix each statement True or False. Justify each answer.
   a. If $Ax = \lambda x$ for some scalar λ , then x is an eigenvector of A.
   b. If $\boldsymbol{v_1}$ ***and*** $\boldsymbol{v_2}$ are linearly independent eigenvectors , then they correspond to distinct eigenvalues.
   c. A steady state vector for a stochastic matrix is actually an eigenvector.
   d. The eigenvalues of a matrix are on its main diagonal .
   e. An eigenspace of A is a null space of certain marix.

6. (5.2.8) Find the characteristic polynomial and the real eigenvalues of the following matrix.

$\begin{bmatrix} -4 & 3 \\ 2 & 1 \end{bmatrix}$

7. (5.2.14) This question requires techniques from Section 3.1. Find the characteristic polynomial of the followig matrix, using either a cofactor expansion or the special formula for 3 x 3 determinants described prior to Exercises 15-18 in section 3.1 [Note : Finding the characteristic polynomial 3 x 3 matrix is not easy to do with just row operations , because the variable λ is involved.]

$$\begin{bmatrix} 4 & 0 & -1 \\ -1 & 0 & 4 \\ 0 & 2 & 3 \end{bmatrix}$$

Diagonalize the matrices in questions 8 and 9, if possible. The real eigenvalues are given below the matrix.

8. (5.3.14)

$$\begin{bmatrix} 2 & 0 & -2 \\ 1 & 3 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

λ=2 , 3

9. (5.3.18)

$$\begin{bmatrix} 2 & -2 & -2 \\ 3 & -3 & -2 \\ 2 & -2 & -2 \end{bmatrix}$$

λ=-2 , -1 , 0

10. (6.1.14) Find the distance between $\boldsymbol{u} = \begin{bmatrix} 0 \\ -5 \\ 2 \end{bmatrix}$ and $\boldsymbol{z} = \begin{bmatrix} -4 \\ -1 \\ 8 \end{bmatrix}$.

Determine which pairs of vectors in questions 11 and 12 are orthogonal.

11. (6.1.16) $\boldsymbol{u} = \begin{bmatrix} 12 \\ 3 \\ -5 \end{bmatrix}, \boldsymbol{v} = \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix}$
12. (6.1.18) $\boldsymbol{y} = \begin{bmatrix} -3 \\ 7 \\ 4 \\ 0 \end{bmatrix}, \boldsymbol{z} = \begin{bmatrix} 1 \\ -8 \\ 15 \\ -7 \end{bmatrix}$
13. (6.1.22) Let $\boldsymbol{u}=(u_1,u_2,u_3)$. Explain why $\boldsymbol{u} \cdot \boldsymbol{u} \geq 0$. When is $\boldsymbol{u} \cdot \boldsymbol{u} = 0$?
14. (6.1.31) Show that if $\boldsymbol{x}$ is in both W and $W^{\perp}$, then $\boldsymbol{x}=0$.
15. (6.2.10) Show that $\{\boldsymbol{u}_1,\boldsymbol{u}_2,\boldsymbol{u}_3\}$ is an ortogonal basis for $\mathbb{R}^3$. The express $\boldsymbol{x}$ a linear combination of the $\boldsymbol{u}$'s.

$\boldsymbol{u_1} = \begin{bmatrix} 3 \\ -3 \\ 0 \end{bmatrix}, \boldsymbol{u_2} = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}, \boldsymbol{u_3} = \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix}$, and $\boldsymbol{x} = \begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}$